University of Dhaka

Department of Computer Science and Engineering

CSE 4113: Internet Programming Lab
4th Year 1st Semester
Session: 2021-2022

Software Requirements Specification (Updated_SRS)

Digital Knowledge Platform (DKP)

Group Name: Semicolon Squad

Submitted by:

Faria Yasmin (21)

Hasibul Islam Sifat (42)

Md. Nuruzzaman (56)

Yuki Bhuiyan (64)

Submitted to:

Md. Fahim Arefin, Lecturer, Department of Computer Science and Engineering, University of Dhakay

Date: 14 April 2026

Contents

1 Document Control **3**
1.1 Metadata . . . 3
1.2 Revision History . . . 3
1.3 Authors and Reviewers . . . 3

2 Problem Definition **4**
2.1 Problem Statement . . . 4
2.2 Target Users . . . 4
2.3 Goals and Non-Goals . . . 5

3 Scope and Context **6**
3.1 In Scope . . . 6
3.2 Out of Scope . . . 7
3.3 Assumptions and Constraints . . . 8

4 Stakeholders and Roles **9**
4.1 Stakeholders . . . 9
4.2 User Roles and Permissions . . . 9

5 Functional Requirements **10**
5.1 MoSCoW Priority Summary . . . 10
5.2 Digital Archive Module . . . 11
5.3 Research Repository Module . . . 13
5.4 Student Project Showcase Module . . . 14
5.5 Digital Library & Catalog Module . . . 15
5.6 Notification & Announcement Center Module . . . 17
5.7 User Account & Profile Management Module . . . 17
5.8 Analytics & Reporting Dashboard Module . . . 18
5.9 Discussion & Comment System Module . . . 19
5.10 Event & Seminar Registry Module . . . 20

6 User Stories and Use Cases **21**
6.1 User Stories . . . 21
6.2 Critical Use Cases . . . 24
6.2.1 Use Case 1: Upload and Publish an Archive Document . . . 24
6.2.2 Use Case 2: Borrow a Book (Lending Workflow) . . . 25
6.2.3 Use Case 3: Student Project Submission and Approval . . . 26
6.2.4 Use Case 4: Advanced Catalog Search by Public Visitor . . . 26
6.2.5 Use Case 5: Bilingual Full-Text Archive Search . . . 27

7 Data Requirements **27**
7.1 Core Entities . . . 28
7.2 Data Dictionary . . . 30
7.3 Data Model . . . 31

8 UI and UX Requirements **34**
8.1 Wireframes . . . 34

8.1.1 Wireframe 1: Public Archive Search & Browse 34
8.1.2 Wireframe 2: Archive Item Detail & Media Viewer 34
8.1.3 Wireframe 3: Student Project Showcase Gallery 35
8.1.4 Wireframe 4: Library Catalog Search (Public) 35
8.1.5 Wireframe 5: Member Dashboard 36
8.1.6 Wireframe 6: Librarian Operations Dashboard 37
8.1.7 Wireframe 7: Archivist Bulk Upload Interface 37
8.1.8 Wireframe 8: Research Output Detail & Citation Export 38
8.1.9 Wireframe 9: Notification Feed 38
8.1.10 Wireframe 10: User Profile Page 39
8.1.11 Wireframe 11: Analytics Dashboard (Admin/Librarian) 39
8.1.12 Wireframe 12: Resource Comments & Discussion 40
8.1.13 Wireframe 13: Event Registry & Calendar 40
8.2 UX and Accessibility Baseline 41

9 Non-Functional Requirements 42

10 Risks and Open Questions 44
10.1 Risk Register 44
10.2 Open Questions 46

11 Testability and Traceability 46
11.1 Requirement Traceability 46
11.2 Initial Manual Test Plan 47

12 Lab 2 Submission Checklist 51

13 Conclusion 51
13.1 Summary 52
13.2 Expected Impact 52
13.3 Team Reflection 53

A Glossary of Terms 53

B Abbreviations 55

C Technology Stack Summary 57

D Sprint Roadmap (Indicative) 58

1 Document Control

1.1 Metadata

Field	Value
Team	Semicolon Squad
Project Name	Digital Knowledge Platform (DKP)
Repository URL	https://github.com/Yukii9291/Digital_Knowledge_platform
Version	0.2 (Revision 2)
Status	Revised
Date	2026-04-14

1.2 Revision History

Version	Date	Author(s)	Change Summary
0.1	2026-03-07	Semicolon Squad	Initial SRS submission (Lab 2).
0.2	2026-04-14	Semicolon Squad	Peer review revision: promoted Bangla metadata search and disaster recovery to Must Have; added restricted-document access request workflow (FR-TXX-007A); clarified testability for FR-TXX-012, FR-TXX-051, FR-TXX-052; added performance benchmarks for NFR-TXX-003; expanded bulk upload edge cases in TC-TXX-008; added AccessRequest entity to data model; added RISK-TXX-003.

1.3 Authors and Reviewers

Role	Name	ID / Email	Date
Author	Faria Yasmin	fariayasmin19@gmail.com	2026-03-07
Author	Hasibul Islam Sifat	hasibulislam-2021211218@cs.du.ac.bd	2026-03-07
Author	Md. Nuruzzaman	md-2021611232@cs.du.ac.bd	2026-03-07
Author	Yuki Bhuiyan	yuki-2021816226@cs.du.ac.bd	2026-03-07

2 Problem Definition

2.1 Problem Statement

What problem are we solving?

Educational institutions in Bangladesh—particularly universities and research labs—manage an enormous volume of academic assets: archived documents, research papers, student project reports, theses, lab portfolios, multimedia resources, and physical library holdings. Today these assets are scattered across:

- **Physical filing cabinets and shelves** with no digital index.
- **Multiple unconnected drives and folders** (Google Drive, Dropbox, local servers) with inconsistent naming, no metadata, and no search capability.
- **Legacy library management systems** that handle only book lending and lack integration with digital resources.
- **No unified student project showcase**, causing project reports to be lost after the semester ends.

This fragmentation leads to:

1. **Discovery failure**—researchers, students, and staff cannot find existing resources, leading to duplicated work.
2. **Knowledge loss**—graduating students' projects, lab experiment records, and institutional memory are not preserved.
3. **Operational inefficiency**—librarians juggle paper ledgers and multiple spreadsheets for lending, returns, and fine tracking.
4. **Access control gaps**—sensitive research data is either over-shared or completely inaccessible.
5. **Language barrier**—existing tools lack Bangla-language search and metadata support, excluding a significant user base.

Why does it matter?

A unified **Digital Knowledge Platform (DKP)** will consolidate all institutional knowledge assets into one searchable, version-controlled, access-managed system. It will preserve institutional memory, accelerate research discovery, streamline library operations, and showcase student talent—ultimately raising the institution's academic profile and operational efficiency.

2.2 Target Users

User Group	Description	Main Needs
Public Visitors	Anyone on the internet visiting the portal	Browse public catalog, view published research, search archives

User Group	Description	Main Needs
Students	Enrolled undergraduate and graduate students	Search/download resources, borrow books, upload project showcases, manage borrowing history
Researchers / Faculty	University faculty and research staff	Upload/manage research outputs, maintain lab portfolios, discover related work
Library Members	Registered borrowers (students + staff)	Borrow/return books, view due dates, manage wishlist, hold requests
Staff / Archivists	Administrative and archive-management staff	Upload/organize documents and media, manage metadata, perform bulk operations
Librarians	Library operations personnel	CRUD on catalog, manage lending/returns, track overdues/fines, generate reports
System Administrators	IT staff maintaining the platform	User management, role/permission configuration, system monitoring, backups

2.3 Goals and Non-Goals

Goals

1. Provide a unified web platform consolidating nine modules: digital archive, research repository, student project showcase, library catalog, notification center, user account management, analytics dashboard, discussion system, and event registry. [**M**]
2. Support single-file document/media uploads (PDF, DOCX) with rich metadata entry [**M**]; extend to multi-format audio/video and in-browser media viewer [**S**].
3. Enable full-text and metadata search in English [**M**]; Bangla metadata search (title, author, tags) [**M**]; Bangla full-text ICU tokenization [**S**].
4. Implement document versioning with lifecycle states: Draft, Review, Published, Archived. [**S**]
5. Enforce four access tiers: Public, Member, Staff, Restricted. [**M**]
6. Deliver a complete lending/return workflow with due-date tracking, overdue notifications [**M**], and fine management [**S**].
7. Provide bulk upload and bulk metadata editing tools for archivists and librarians. [**C**]
8. Offer member account features: borrowing history, wishlist [**S**]; hold requests [**C**].
9. Generate operational reports: circulation, overdue, most-borrowed statistics. [**S**]
10. Provide secure user registration and authentication via email/password and OAuth (Google, institutional SSO) with JWT-based session management. [**M**]
11. Maintain per-user profiles with unified activity history, notification preferences, and self-service account management. [**M/S**]

12. Deliver a real-time in-app notification feed and configurable email digests for due-date reminders, uploads, approvals, and announcements. [**S**]
13. Offer platform-wide analytics dashboards to administrators and librarians with usage statistics, engagement metrics, and exportable reports. [**S**]
14. Enable community engagement through threaded comments and staff-moderated content. [**S**]
15. Support academic event management with event listings, material attachments, member RSVP with capacity enforcement, and a calendar view. [**S**]

Non-Goals

1. **E-commerce / payment gateway integration**—fines are tracked but not collected online in v1.
2. **AI-powered recommendation engine**—out of scope for the initial release.
3. **Mobile-native applications**—the platform will be responsive web; dedicated iOS/Android apps are deferred.
4. **Inter-library loan across institutions**—only intra-institutional lending is supported.
5. **Plagiarism detection**—external tools may integrate later but are not part of this SRS.
6. **Real-time collaborative editing**—documents are uploaded as files, not edited in-browser.

3 Scope and Context

3.1 In Scope

Module	Features
Digital Archive	[M] Single-file document upload (PDF, DOCX, PPTX, TXT), rich metadata editor (English title required; Bangla optional), access tier enforcement, English full-text + metadata search, browse and filter; [S] multi-format media upload (images, audio, video), in-browser media viewer, document versioning lifecycle (draft/review/published/archived), audit trail; [C] bulk upload (up to 50 files), bulk metadata editing
Research Repository	[M] Research output upload pages with structured metadata (title, abstract, authors, keywords, DOI), public-facing research listing; [S] author profile pages, citation export (BibTeX, APA, MLA), persistent identifiers; [C] lab portfolio management (lab pages with members and outputs)

Module	Features
Student Project Showcase	[M] Project submission with metadata, advisor approval workflow, public showcase gallery, project detail pages; [S] filtering by department/year/technology, project detail pages with embedded media
Digital Library & Catalog	[M] Advanced catalog search with faceted filters, item CRUD for librarians, lending/return workflow with manual ID entry, due-date tracking and overdue notifications; [S] fine management, member borrowing history, wishlist, librarian dashboard and reports; [C] hold requests queue, CSV/Excel import/export; [W] barcode/QR scanning (excluded in v1)
Notification & Announcement Center	[M] Admin broadcast announcements, in-app notification feed with unread count; [S] email digest (daily/weekly) for due-date reminders and new uploads, per-user notification preference settings
User Account & Profile Management	[M] Email/password and OAuth (Google, institutional SSO) registration and login, user profile page (avatar, bio, department, role badges), account settings (password change, notification preferences, personal data export); [S] unified activity history, admin-initiated or self-service account deactivation
Analytics & Reporting Dashboard	[S] Platform-wide usage statistics (uploads, downloads, searches over time), most-accessed resources, user engagement metrics, search analytics (English; Bangla metadata search is **M** per FR-TXX-004), exportable PDF/CSV reports for any metric and date range
Discussion & Comment System	[S] Threaded comments on archive items, research papers, and projects; moderation tools (hide, delete, flag); [C] Q&A on resources with author/admin answers; reactions (Like, Helpful, Insightful)
Event & Seminar Registry	[S] Staff-created event listings with date, venue and description; event materials attachment (slides, recordings); member RSVP with capacity enforcement; past events archive; [C] monthly calendar view of upcoming events

3.2 Out of Scope

- Native mobile applications (iOS / Android).
- Online payment processing for fines.
- Multi-institutional federation or inter-library loan.
- Plagiarism detection or AI-based content analysis.
- Real-time in-browser document editing.

- Integration with external Learning Management Systems (Moodle, Canvas) in v1.

3.3 Assumptions and Constraints

Technical Constraints

- The platform will be built as a web application accessible via modern browsers (Chrome, Firefox, Edge, Safari).
- Backend: Node.js (Express) or fastapi; Frontend: React.js or Next.js.
- Database: PostgreSQL for relational data; Elasticsearch for full-text search (English — **Must Have**; Bangla metadata search — **Must Have**; Bangla full-text ICU analyzer — **Should Have**).
- File storage: server-side file storage on the application server's local filesystem or network-attached storage.
- Maximum single-file upload size: 500 MB.
- The system must support at least 500 concurrent users.

Organizational Constraints

- The development team consists of 4–5 members with a 14-week semester timeline.
- Access to institutional branding assets and sample data is required.
- Librarian input is needed for catalog field definitions and lending rules.

Timeline Constraints

- SRS completion
- SRS Review
- SDD
- Peer UI critique
- Testing baseline
- Peer QA
- Submission

4 Stakeholders and Roles

4.1 Stakeholders

Stakeholder	Interest	Priority
University Administration	Institutional knowledge preservation, public image	High
Library Department	Streamlined operations, reduced manual effort	High
Faculty / Researchers	Research discoverability, lab portfolio visibility	High
Students	Easy access to resources, project showcase, borrowing	High
IT Department	Maintainability, security, infrastructure costs	Medium
Public / Prospective Students	Access to public resources, institutional credibility	Medium
Course Instructor	Lab deliverable compliance, grading	High

4.2 User Roles and Permissions

Role	Description	Permissions
Guest	Unauthenticated public visitor	View public catalog, browse published archives/research/showcase, search
Member	Registered library member (student/staff)	All Guest permissions + borrow/return [M], borrowing history [S], wishlist [S], hold requests [C], download member-tier resources [M], manage own profile and notification preferences, post comments and reactions, RSVP for events
Student Author	Student submitting a project	All Member permissions + create/edit own project submissions, upload project media
Researcher	Faculty or research staff	All Member permissions + upload/manage research outputs, maintain lab portfolio, edit own research metadata
Archivist / Staff	Archive management personnel	All Member permissions + upload/edit/delete archive items [M], manage access tiers for items [M]; bulk upload and bulk metadata editing [C]

Role	Description	Permissions
Librarian	Library operations staff	All Staff permissions + CRUD on catalog items [M], manage lending/returns [M], track overdues [M], manage fines [S], generate reports [S]; import/export catalog data [C]; view analytics dashboard [S]; export analytics reports [S]
Admin	System administrator	All permissions + user management, role assignment, system configuration, audit logs, view platform analytics, broadcast announcements, create/manage events, moderate comments and Q&A

5 Functional Requirements

Requirement ID format: `FR-TXX-###`

5.1 MoSCoW Priority Summary

Code	Label	Scope for This Project
M	Must Have (Core MVP)	Single-file document uploads (PDF/DOCX) with rich metadata; RBAC access tiers (Public/Member/Staff/Restricted); research output and student project showcase pages; librarian catalog CRUD; English-language archive and catalog search with category filters; Bangla metadata search (title, author, tags); restricted-document access request workflow; lending/return workflow with due-date tracking; user authentication and profile management; disaster recovery with automated failover.
S	Should Have (Portfolio Enhancers)	Multi-format media uploads (images, audio, video) and in-browser media viewer; document versioning lifecycle (draft/review/published/archived); overdue fine calculation; member borrowing history and wishlist; librarian dashboard and PDF/CSV reports; email digest notifications; audit trail; Bangla full-text ICU tokenization.

Code	Label	Scope for This Project
C	Could Have (Stretch Goals)	Lab portfolio pages; book hold request queue; bulk upload and bulk metadata editing; CSV/Excel catalog import/export; loan renewal.

5.2 Digital Archive Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-001	Document Upload	The system shall allow authorized users to upload documents in PDF, DOCX, PPTX, and TXT formats up to 500 MB per file.	M	File uploads successfully; unsupported formats rejected with error.
FR-TXX-002	Media Upload	The system shall support upload of images (JPEG, PNG, TIFF, WebP), audio (MP3, WAV, OGG), and video (MP4, WebM, AVI).	S	Each format uploads and previews correctly.
FR-TXX-003	Rich Metadata Editor	The system shall provide a metadata editor: title (Bangla + English), description, author(s), date, tags, category, language, access tier, custom fields.	M	All fields editable and saved; Bangla text renders correctly.
FR-TXX-004	Bilingual Search	The system shall provide full-text and metadata search in Bangla and English with Bangla tokenization. Bangla metadata search (title, author, tags) is mandatory; full-text ICU tokenization is deferred to a later sprint.	M	A Bangla keyword matching a document title or description appears in the top 5 results; English queries return results ranked by relevance score.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-005	Browse & Filter	The system shall allow browsing with filters: category, date range, media type, language, access tier, tags.	M	Filters narrow results correctly; combinable; clearable.
FR-TXX-006	Document Versioning	The system shall maintain version history with states: Draft, Review, Published, Archived.	S	State transitions follow workflow; previous versions accessible.
FR-TXX-007	Access Tier Enforcement	The system shall enforce four access tiers (Public, Member, Staff, Restricted) per item.	M	Unauthorized access is blocked with appropriate message.
FR-TXX-007A	Restricted Access Request	The system shall allow members to submit a request (with reason) for access to Restricted-tier documents. The archivist/admin receives a notification, reviews the request, and approves or denies it. The requesting member is notified of the outcome.	M	Member submits request with reason; archivist sees pending request; approve grants access; deny returns rejection message to member.
FR-TXX-008	Media Viewer	The system shall provide in-browser viewer for PDFs, images, audio, and video.	S	All formats render without plugin installation.
FR-TXX-009	Bulk Upload	The system shall allow batch upload of up to 50 files with progress indicators.	C	50 files upload; progress shown per file; failures listed.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-010	Bulk Metadata Edit	The system shall allow selecting multiple items and applying metadata changes in bulk.	C	Multi-select works; changes apply to all; confirmation shown.
FR-TXX-011	Download with Access Check	The system shall allow authorized users to download original files based on access tier.	M	Authorized download succeeds; unauthorized see "Access Denied."
FR-TXX-012	Audit Trail	The system shall log all upload, edit, delete, and access events with timestamp, user ID, action. Completeness is verified for a representative sample of 5 action types (upload, edit metadata, delete, download, status change); remaining actions verified via code review.	S	Representative sample logs created and verifiable; admins can view filtered audit logs.

5.3 Research Repository Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-013	Research Output Upload	Allow researchers to upload papers, reports, datasets with structured metadata (title, abstract, authors, keywords, DOI).	M	Upload succeeds; metadata saved; linked to researcher profile.
FR-TXX-014	Author Profile Page	Generate author profile listing all outputs, lab affiliations, contact info.	S	Profile shows name, affiliation, publications sorted by date.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-015	Lab Portfolio	Allow research groups to create lab pages with description, members, projects, outputs.	C	Lab page created; members linked; outputs listed.
FR-TXX-016	Citation Export	Provide citation export in BibTeX, APA, and MLA formats.	C	Each format generates correct text; .bib file is valid.
FR-TXX-017	Research Search	Search research outputs by title, author, keyword, year, type.	M	Combined queries narrow results correctly.
FR-TXX-018	Persistent Identifier	Assign unique persistent identifier to each published output.	S	Identifier displayed on item page and in citation exports.

5.4 Student Project Showcase Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-019	Project Submission	Allow students to submit projects with title, abstract, team, advisor, semester, department, technologies, media.	M	All fields validated; files upload; saved as Draft.
FR-TXX-020	Advisor Approval	Route submissions to advisor for review and approval before public visibility.	M	Advisor notified; can Approve or Request Changes.
FR-TXX-021	Public Gallery	Display approved projects in browsable gallery with thumbnail, title, team, semester.	M	Gallery loads with pagination; cards show correct data.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-022	Project Detail Page	Display full details with embedded viewer, video, source code links, team profiles.	M	All media renders; links open in new tabs.
FR-TXX-023	Showcase Filters	Filter projects by department, semester, technology tags, advisor.	S	Filters combinable; URL reflects state for shareability.
FR-TXX-024	Project Edit	Allow student authors to edit details until final approval.	S	Edit pre-fills data; after approval, editing is locked.

5.5 Digital Library & Catalog Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-025	Catalog CRUD	Librarians can create, read, update, delete catalog items with title, ISBN, author, publisher, copies, shelf location.	M	All CRUD operations work; soft-delete enabled.
FR-TXX-026	Advanced Search	Faceted search: title, author, ISBN, category, availability, year range, media type.	M	Each facet narrows results; availability shown.
FR-TXX-027	Public Portal	Public search interface; unauthenticated users can view details but not borrow.	M	Borrow button hidden for guests.
FR-TXX-028	Lending Workflow	Issue a book to a member recording borrower ID, issue date, due date.	M	Transaction recorded; copies decremented; dashboard updated.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-029	Return Workflow	Process returns, update availability, calculate overdue fines.	M	Return recorded; copies incremented; fine calculated if overdue.
FR-TXX-030	Due Date Tracking	Display due dates; send reminders 3 days before and on due date.	M	Reminders sent; overdue items highlighted.
FR-TXX-031	Overdue & Fines	Auto-calculate fines; librarians can waive or adjust.	S	Fines correct per day; waive/adjust reflected on account.
FR-TXX-032	Barcode/QR Gen	Generate barcode and QR labels, downloadable as printable PDF.	W	Codes generated; scannable; PDF print-ready.
FR-TXX-033	Barcode/QR Scan	Support scanning via webcam or handheld to look up items.	W	Scan identifies correct item; detail page opens.
FR-TXX-034	Borrowing History	Display member's complete borrowing history with dates, status, fines.	S	Chronological; shows all fields correctly.
FR-TXX-035	Wishlist	Members can add items to personal wishlist.	S	Add/remove works; persisted across sessions.
FR-TXX-036	Hold Requests	Members place holds on unavailable items; notified when available.	C	Hold placed; FIFO order; notification sent on availability.
FR-TXX-037	Librarian Dash-board	Dashboard: on-loan, overdue, holds, daily transactions, statistics.	S	Loads within 3s; all widgets correct.
FR-TXX-038	Reports	Circulation, overdue, most-borrowed, growth, activity reports as PDF/CSV.	S	Reports generate correctly; date filtering works.
FR-TXX-039	Import/Export	Import catalog from CSV/Excel; export full catalog.	C	Import parses correctly; duplicates detected; export complete.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-040	Renewal	Members renew once (if no holds) extending due date by loan period.	C	Renewal extends date; second attempt rejected; blocked if hold exists.

5.6 Notification & Announcement Center Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-041	System Announcements	Admins shall broadcast announcements to all users or specific role groups.	M	Announcement created; target group receives it in-app and by email.
FR-TXX-042	In-App Notification Feed	The system shall display a per-user notification feed with unread count badge and full history.	M	Bell icon updates in real time; history lists all events chronologically.
FR-TXX-043	Email Digest	The system shall send configurable daily or weekly email digests covering due-date reminders, new uploads, and approvals.	S	Digest emails sent on schedule; content reflects real events; unsubscribe link present.
FR-TXX-044	Notification Preferences	Users shall choose notification channels (in-app, email) and event types they subscribe to.	S	Preferences saved; only selected events trigger notifications; changes immediate.

5.7 User Account & Profile Management Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-045	Registration & Login	The system shall support email/password registration and OAuth login (Google, institutional SSO).	M	Both methods authenticate; JWT issued; invalid credentials rejected.
FR-TXX-046	User Profile Page	Each user shall have a profile page with avatar, bio, department, role badges, and activity summary.	M	Profile renders correctly; avatar upload works; bio editable up to 500 characters.
FR-TXX-047	Account Settings	Users shall change password, manage notification preferences, and export personal data.	M	Password change requires current password; export generates JSON/CSV of user data.
FR-TXX-048	Activity History	The system shall show a unified per-user log of downloads, borrows, uploads, submissions, and comments.	S	Log is chronological; filterable by type; shows timestamps and resource names.
FR-TXX-049	Account Deactivation	Users may request self-service deactivation; admins may force-deactivate any account.	S	Deactivated accounts cannot log in; data retained (soft-delete); admin can reactivate.

5.8 Analytics & Reporting Dashboard Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-050	Usage Statistics	The system shall display platform-wide counts of uploads, downloads, and searches by day, week, and month.	S	Charts render correctly; date range selectable; data matches backend logs.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-051	Most Accessed Resources	The system shall list the top 10 catalog items by total download count over the past 30 calendar days. Views without download are counted separately. Counts are unique per user per day.	S	Top 10 list matches expected ranking from seeded data; clicking an item opens its detail page; list refreshes daily.
FR-TXX-052	User Engagement Metrics	The system shall report active users, new registrations, and returning visitor rates over selectable periods.	S	Verified by seeding the analytics event log with synthetic events via a test script and confirming dashboard figures match the seeded counts. Metrics update daily; graphs display trends.
FR-TXX-053	Search Analytics	The system shall record most common search terms (Bangla & English) and zero-result queries.	S	Top 20 terms listed; zero-result list sortable; exportable as CSV.
FR-TXX-054	Analytics Export	Admins and librarians shall export any analytics report as PDF or CSV for a chosen date range.	S	Exported file contains correct data; date filter works; large exports complete within 30 s.

5.9 Discussion & Comment System Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-055	Threaded Comments	Authenticated members shall post and reply to comments on archive items, research papers, and projects.	S	Comments thread correctly; timestamps shown; author name linked to profile.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-056	Q&A on Resources	Members shall post questions on resource pages; authors and admins can mark an answer as accepted.	S	Questions appear separately from comments; accepted answer highlighted.
FR-TXX-057	Comment Moderation	Staff and admins shall hide, delete, or flag comments for review.	S	Moderation actions take effect immediately; flagged items appear in moderation queue.
FR-TXX-058	Reactions	Members shall add reactions (Like, Helpful, Insightful) to items and comments; one reaction per item per user.	C	Reaction counts update in real time; removable by the same user.

5.10 Event & Seminar Registry Module

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-059	Event Creation	Staff shall create academic event listings (seminars, workshops, lab open days) with title, date, venue, and description.	S	Event saved; visible immediately; required fields validated.
FR-TXX-060	Event Materials	Staff shall attach slides, recordings, and reports to an event entry.	S	Files attached and downloadable by members; access tier per file enforced.
FR-TXX-061	RSVP / Registration	Members shall RSVP for events; system enforces a maximum capacity limit.	S	RSVP recorded; capacity counter updated; over-capacity requests rejected with waitlist option.
FR-TXX-062	Past Events Archive	All concluded events and their materials shall remain permanently browsable.	S	Past events searchable; materials downloadable; clearly labelled as concluded.

Req ID	Feature	Requirement Statement	Pri.	Acceptance Criteria
FR-TXX-063	Calendar View	The system shall display upcoming events in a monthly calendar view with navigation between months.	S	Calendar renders by month; clicking a date shows events; navigation works.

6 User Stories and Use Cases

6.1 User Stories

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-TXX-001	Public Visitor	to search the archive by keyword in Bangla or English	I can find documents regardless of the script I type in	Bangla and English queries both return relevant results; titles display correctly in the respective script
US-TXX-002	Archivist	to upload a batch of 30 scanned documents with metadata	I can digitize an entire folder efficiently	All 30 files upload; metadata saved; failures show error
US-TXX-003	Researcher	to upload my published paper with DOI and author info	it is discoverable and properly cited	Paper in search; BibTeX correct; author profile updated
US-TXX-004	Student	to submit my final-year project with report and demo video	my work is showcased after advisor approval	Report + video accepted; advisor notified; gallery shows after approval
US-TXX-005	Advisor	to review and approve a student project	only quality work is published	Pending queue visible; approve/reject works; approved goes live
US-TXX-006	Library Member	to search the catalog and borrow a book	I can access physical resources	Search finds book; availability shown; borrow recorded; dashboard updated

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-TXX-007	Library Member	to view my borrowing history and fines	I can track activity and clear dues	History shows all borrows; fines correct; total balance displayed
US-TXX-008	Librarian	to manually enter a book ID and member ID to process a return	I can handle returns quickly without hardware dependencies	Book ID and member ID entry accepted; return processed; fine auto-calculated if overdue
US-TXX-009	Librarian	to generate a monthly circulation report	I can report usage to administration	Report for date range; total issues/returns/overdues; PDF/CSV export
US-TXX-010	Library Member	to place a hold on a borrowed book	I am notified when available	Hold placed; queue position shown; notification sent on return
US-TXX-011	Admin	to manage user roles and permissions	proper access control is maintained	Assign/revoke roles; changes immediate; audit log records
US-TXX-012	Archivist	to change access tier of 20 items to Public	declassified documents become public	Multi-select; change tier; confirm; all 20 now public
US-TXX-013	Public Visitor	to browse showcase by department	I can explore projects from a specific field	Department filter works; count updates; correct results
US-TXX-014	Researcher	to create a lab portfolio page	our work is visible in one place	Lab page with description, members, publications; URL shareable
US-TXX-015	Library Member	to add a book to my wishlist	I can remember to borrow it later	Add/remove works; wishlist persisted; page lists all items
US-TXX-016	Librarian	to import 500 catalog records from CSV	I can migrate data quickly	500 rows parsed; errors per row; valid imported; summary shown

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-TXX-017	Student	to search and download research papers	I can access references for my thesis	Search returns papers; download works within access tier
US-TXX-018	Archivist	to view version history and restore a previous version	I can recover from mistakes	History shows all versions; restore creates new version from selected
US-TXX-019	Admin	to send a system-wide announcement to all users	I can communicate important updates instantly	Announcement created; all users receive in-app notification and email
US-TXX-020	Member	to set my notification preferences	I only receive alerts relevant to me	Preferences saved; only chosen event types and channels trigger notifications
US-TXX-021	New User	to register using my Google account	I can sign up without creating a new password	OAuth flow completes; profile created; role assigned; redirected to dashboard
US-TXX-022	Member	to view my unified activity history	I can track everything I have done on the platform	Log shows borrows, downloads, and uploads in chronological order
US-TXX-023	Admin	to view platform-wide upload and download statistics	I can monitor usage trends	Dashboard shows daily/weekly/monthly charts; data matches backend logs
US-TXX-024	Librarian	to see the most-borrowed books and zero-result search terms	I can improve the collection based on demand	Top 10 list and zero-result queries displayed; exportable as CSV
US-TXX-025	Member	to post a question on a research paper page	I can get clarification from the author	Question posted; author notified; accepted answer highlighted on page

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-TXX-026	Staff	to hide an inappropriate comment	community standards are maintained	Comment hidden immediately; appears in moderation queue for review
US-TXX-027	Staff	to create a seminar event with slides attached	attendees can access materials after the event	Event saved with files; appears in calendar and past events archive
US-TXX-028	Member	to RSVP for an upcoming workshop	I secure my place before capacity is reached	RSVP recorded; capacity counter decremented; confirmation notification sent
US-TXX-029	Member	to request access to a Restricted-tier document by providing a reason	I can obtain permission without contacting the admin outside the platform	Request submitted with reason; archivist notified; on approval, document becomes accessible; on denial, rejection message displayed

6.2 Critical Use Cases

6.2.1 Use Case 1: Upload and Publish an Archive Document

Preconditions:

- User is authenticated with Archivist or Librarian role.
- File is in a supported format and $\leq 500\,\text{MB}$.

Main Flow:

1. User navigates to Archive → Upload.
2. User selects file(s) from local filesystem.
3. System validates file type and size.
4. Upload progress bar is displayed.
5. On completion, metadata editor form is presented pre-populated with auto-extracted data.
6. User fills in: title (Bangla + English), description, author(s), tags, category, language, access tier.
7. User clicks "Save as Draft."
8. System saves the item with status = Draft.
9. User (or reviewer) changes status to "Review."
10. Reviewer examines the item and changes status to "Published."

11. Item becomes discoverable in search according to its access tier.

Alternate / Exception Flows:

- **3a.** Unsupported format → Error message; upload cancelled.
- **3b.** File exceeds 500 MB → Size error; upload cancelled.
- **4a.** Network interruption → "Upload failed. Please retry."
- **10a.** Reviewer requests changes → Status returns to Draft; author notified.

Postconditions:

- Document stored in server-side file storage.
- Metadata indexed in search engine.
- Version 1.0 recorded.
- Audit log entry created.

6.2.2 Use Case 2: Borrow a Book (Lending Workflow)

Preconditions:

- Member is authenticated with active membership.
- No overdue books; fines below blocking threshold.
- Desired book has at least one available copy.

Main Flow:

1. Member searches for the book in the catalog.
2. System displays results with availability status.
3. Member selects the book and views detail page.
4. Member approaches librarian desk with physical book.
5. Librarian manually enters the book's catalog ID into the lending form.
6. System identifies the catalog item.
7. Librarian enters the member's ID.
8. System verifies member eligibility.
9. Librarian confirms the issue.
10. System records: borrower ID, item ID, issue date, due date.
11. Available copies decremented by 1.
12. Member dashboard shows the new active borrow.

Alternate / Exception Flows:

- **8a.** Member has overdue books → Warning displayed.
- **8b.** Borrow limit reached → "Borrow limit reached (max 5 items)."
- **6a.** Catalog ID not found → "No item found. Please check the ID and retry."
- **10a.** No copies available (race condition) → Offer hold.

Postconditions:

- Lending transaction recorded.
- Member's active borrow list updated.
- Catalog availability decremented.
- Audit log records the transaction.

6.2.3 Use Case 3: Student Project Submission and Approval

Preconditions:

- Student is authenticated with Student Author role.
- Active submission period is open.

Main Flow:

1. Student navigates to Showcase → Submit Project.
2. Student fills in: title, abstract, team members, advisor, semester, department, technologies.
3. Student uploads project report (PDF), optionally demo video (MP4) and source code link.
4. System validates required fields and file formats.
5. Student clicks "Submit for Review."
6. System saves with status = Pending Review.
7. Advisor receives email notification.
8. Advisor reviews submission.
9. Advisor clicks "Approve" (or "Request Changes").
10. If approved, project appears in public showcase gallery.

Alternate / Exception Flows:

- **4a.** Required field missing → Validation error highlighted.
- **9a.** Advisor requests changes → Student notified with comments.
- **3a.** Video exceeds 500 MB → Size error; upload cancelled.

Postconditions:

- Approved project publicly visible.
- Student's profile lists the project.
- Advisor's profile shows supervised projects.

6.2.4 Use Case 4: Advanced Catalog Search by Public Visitor

Preconditions:

- Catalog contains at least one item.
- No authentication required.

Main Flow:

1. Visitor navigates to the public catalog portal.
2. Visitor enters a keyword in English (e.g., "data structures") or in Bangla (e.g., "*data structures*" in Bangla script).
3. System detects the input script and routes the query through the appropriate language analyzer (standard for English; ICU Bangla for Bangla).
4. System returns results with faceted filters regardless of the input language.
5. Visitor applies filters: Category = "Textbook," Year = "2024–2026."
6. System narrows results.
7. Visitor clicks a result to view details.
8. Visitor notes the shelf location.

Alternate / Exception Flows:

- **4a.** No results for English query → "No results found. Try different keywords."
- **4b.** No results for Bangla query → "No results found. Try different keywords (Bangla or English)."
- **7a.** Visitor clicks "Borrow" → Prompt "Please log in."

Postconditions:

- Search query (in either language) reflected in URL for shareability.
- No data modified.

6.2.5 Use Case 5: Bilingual Full-Text Archive Search

Preconditions:

- Archive contains at least one indexed document.
- Search engine is configured with both a standard English analyzer and a Bangla ICU analyzer.

Main Flow (English Search):

1. User navigates to the Archive search page.
2. User types an English keyword (e.g., "machine learning") into the search bar.
3. System identifies the query as English (Latin script) and applies the standard English analyzer.
4. System returns ranked results matching English titles, descriptions, and tags.
5. User browses results; both English and Bangla titles are displayed for each item.

Main Flow (Bangla Search):

1. User navigates to the Archive search page.
2. User types a Bangla keyword (e.g., in Bangla script) into the search bar.
3. System identifies the query as Bangla (Unicode block U+0980–U+09FF) and applies the ICU Bangla analyzer.
4. System returns ranked results matching Bangla titles, descriptions, and tags.
5. User browses results; both Bangla and English titles are displayed for each item.

Alternate / Exception Flows:

- **3a.** Mixed-script query (Bangla + English) → System runs parallel queries with both analyzers and merges the result sets by relevance score.
- **4a.** No results in either language → "No matching documents found. You may search in Bangla or English."
- **2a.** Empty query submitted → Validation error: "Please enter a keyword to search."

Postconditions:

- Search term (in either script) is stored for analytics (FR-TXX-053).
- Search query reflected in URL for shareability.
- No data is created, modified, or deleted.

7 Data Requirements

7.1 Core Entities

Entity	Description	Key Fields	Constraints
User	All platform users	user_id, name, email, password_hash, role, department, membership_status	email unique; role is enum
ArchiveItem	Digital archive document/media	item_id, title_en, title_bn, description, file_url, file_type, file_size, access_tier, status, version	title_en required; access_tier enum
ArchiveVersion	Version history	version_id, item_id (FK), version_number, file_url, metadata_snapshot, changed_by	item_id refs ArchiveItem
Tag	Categorization tags	tag_id, name_en, name_bn	name_en unique
ResearchOutput	Research papers, datasets	output_id, title, abstract, authors (JSON), keywords, doi, file_url, lab_id (FK)	title required
Lab	Research group	lab_id, name, description, head_researcher (FK)	name unique
StudentProject	Project submission	project_id, title, abstract, team_members (JSON), advisor_id (FK), semester, department, technologies	title required; advisor_id refs User
CatalogItem	Library item	catalog_id, title, isbn, authors, publisher, edition, year, category, total_copies, available_copies, shelf_location, barcode	isbn unique (nullable)

Entity	Description	Key Fields	Constraints
LendingTransaction	Borrow/return record	transaction_id, catalog_id (FK), member_id (FK), issue_date, due_date, return_date, fine_amount	FKs reference CatalogItem, User
HoldRequest	Hold queue	hold_id, catalog_id (FK), member_id (FK), request_date, status	FIFO by request_date
Wishlist	Saved items	wishlist_id, member_id (FK), catalog_id (FK), added_at	Unique (member_id, catalog_id)
Fine	Fine tracking	fine_id, member_id (FK), transaction_id (FK), amount, reason, status	amount ≥ 0
AuditLog	Audit trail	log_id, user_id (FK), action, entity_type, entity_id, details (JSON), timestamp	Append-only; immutable
Notification	User notifications	notification_id, user_id (FK), type, message, read, created_at	type is enum
Announcement	Admin broadcast message	announcement_id, created_by (FK), title, body, target_role (nullable), created_at	null target_role broadcasts to all users
Comment	Threaded comment or Q&A on a resource	comment_id, user_id (FK), entity_type, entity_id, parent_id (FK nullable), body, is_answer, hidden, created_at	parent_id allows threading; entity_type is enum (archive, research, project)

Entity	Description	Key Fields	Constraints
Reaction	User reaction on an item or comment	reaction_id, user_id (FK), entity_type, entity_id, reaction_type	Unique (user_id, entity_type, entity_id); reaction_type: like, helpful, insightful
Event	Academic event listing	event_id, title, description, event_date, venue, capacity (nullable), created_by (FK), status	status: upcoming, cancelled, concluded; null capacity means unlimited
EventMaterial	File attached to an event	material_id, event_id (FK), file_url, file_type, title, access_tier	event_id refs Event; access_tier enforced on download
EventRSVP	Member registration for an event	rsvp_id, event_id (FK), member_id (FK), rsvp_at, waitlisted	Unique (event_id, member_id); waitlisted flag set when over capacity
AccessRequest	Request for Restricted-tier item access	request_id, user_id (FK), item_id (FK), reason (TEXT), status, reviewed_by (FK nullable), reviewed_at	status enum: pending, approved, denied; user_id refs User; item_id refs ArchiveItem

7.2 Data Dictionary

Field	Type	Req.	Validation	Example
user_id	UUID	Yes	Auto-generated	a1b2c3d4-e5f6...
email	VARCHAR(255)	Yes	Valid email; unique	student@uni.edu.bd
password_hash	VARCHAR(255)	Yes	Bcrypt hash; min 8 char input	$2b$12$...
role	ENUM	Yes	Allowed role values	member
title_en	VARCHAR(500)	Yes	Non-empty; max 500	Introduction to Algorithms
title_bn	VARCHAR(500)	No	Max 500; Bangla Unicode	(Bangla text)
file_url	TEXT	Yes	Valid file path / URL	/storage/uploads/doc.pdf

Field	Type	Req.	Validation	Example
file_type	VARCHAR(20)	Yes	Allowed MIME types	application/pdf
file_size	BIGINT	Yes	> 0, ≤ 524288000	10485760
access_tier	ENUM	Yes	public, member, staff, restricted	public
status	ENUM	Yes	draft, review, published, archived	published
isbn	VARCHAR(17)	No	Valid ISBN-10 or ISBN-13	978-0-262-03384-8
total_copies	INTEGER	Yes	≥ 1	5
available_copies	INTEGER	Yes	≥ 0 and $\leq$ total_copies	3
issue_date	DATE	Yes	$\leq$ current date	2026-03-07
due_date	DATE	Yes	$>$ issue_date	2026-03-21
fine_amount	DECIMAL(10,2)	Yes	≥ 0.00	25.00
barcode	VARCHAR(50)	No	Unique if present	DKP-CAT-000123
semester	VARCHAR(20)	Yes	Pattern: Season YYYY	Spring 2026
technologies	JSONB	No	Array of strings	["React","Node.js"]
comment_body	TEXT	Yes	Non-empty; max 2000 characters	Great resource; this helped my research.
reaction_type	ENUM	Yes	like, helpful, insightful	helpful
event_date	DATETIME	Yes	ISO 8601; must be a future date on creation	2026-04-15T14:00:00+06:00
capacity	INTEGER	No	≥ 1 if provided; null means unlimited	50
target_role	VARCHAR(20)	No	Valid role value or null (= all users)	member

7.3 Data Model

The Entity-Relationship diagram below shows the major relationships. *Attach a visual ER diagram (Figma/draw.io) for the final submission.*

```
+----------+          +--------------+          +----------------+
```

```
|   User   |--1:N--| ArchiveItem  |--1:N--| ArchiveVersion |
| user_id  |       | item_id      |       | version_id     |
| name     |       | title_en/bn  |       | version_num    |
| email    |       | file_url     |       | file_url       |
| role     |       | access_tier  |       | changed_by     |
+----------+       | status       |       +----------------+
     |              +------+-------+
     |                     | M:N
     |              +------+-------+
     |              |     Tag      |
     |              | tag_id       |
     |              | name_en/bn   |
     |              +--------------+
     |
     +--1:N--+------------------+
     |       | ResearchOutput   |--N:1--+----------+
     |       | output_id        |       |   Lab    |
     |       | title, abstract  |       | lab_id   |
     |       | doi, file_url    |       | name     |
     |       +------------------+       +----------+
     |
     +--1:N--+------------------+
     |       | StudentProject   |
     |       | project_id       |
     |       | title, abstract  |
     |       | advisor_id (FK)  |
     |       +------------------+
     |
     +--1:N--+---------------------+     +--------------+
     |       | LendingTransaction  |--N:1| CatalogItem  |
     |       | transaction_id      |     | catalog_id   |
     |       | issue/due/return    |     | title, isbn  |
     |       | fine_amount         |     | copies       |
     |       +---------------------+     | barcode/qr   |
     |                                   +------+-------+
     +--1:N--+------------------+               |
     |       |  HoldRequest     |--N:1----------+
     |       +------------------+
     |
     +--1:N--+------------------+
     |       |   Wishlist       |--N:1-- CatalogItem
     |       +------------------+
     |
     +--1:N--+------------------+
     |       |     Fine         |--N:1-- LendingTransaction
     |       +------------------+
     |
     +--1:N-- AuditLog
```

```
     +--1:N-- Notification
     +--1:N-- Announcement (created)
     +--1:N-- Comment (posted)
     +--1:N-- Reaction (made)
     +--1:N-- Event (created)
     +--1:N-- EventRSVP
     +--1:N-- AccessRequest (submitted)

+-------------------+    +-----------------+    +------------------+
|      Event        |-1:N-| EventMaterial   |    |   EventRSVP      |
| event_id          |    | material_id     |    | rsvp_id          |
| title, date       |    | file_url, title |    | event_id (FK)    |
| venue, capacity   |    | access_tier     |    | member_id (FK)   |
| status            |    +-----------------+    | waitlisted       |
+-------------------+                           +------------------+

+-------------------+    +---------------------------+
|     Comment       |    |        Reaction           |
| comment_id        |    | reaction_id               |
| user_id (FK)      |    | user_id (FK)              |
| entity_type       |    | entity_type, entity_id    |
| entity_id         |    | reaction_type             |
| parent_id (FK?)   |    +---------------------------+
| body, hidden      |
+-------------------+

+---------------------------+
|      AccessRequest        |
| request_id                |
| user_id (FK) --> User     |
| item_id (FK) --> ArchiveItem |
| reason (TEXT)             |
| status: pending/approved/ |
|         denied            |
| reviewed_by (FK nullable) |
| reviewed_at               |
+---------------------------+
```

Major Relationships:

- **User** ↔ **ArchiveItem**: One user uploads many items (1:N).
- **ArchiveItem** ↔ **ArchiveVersion**: One item has many versions (1:N).
- **ArchiveItem** ↔ **Tag**: Many-to-many via junction table.
- **User** ↔ **ResearchOutput**: One researcher uploads many outputs (1:N).
- **ResearchOutput** ↔ **Lab**: Many outputs belong to one lab (N:1).
- **User** ↔ **LendingTransaction** ↔ **CatalogItem**: Member borrows items.
- **CatalogItem** ↔ **HoldRequest**: One item has many holds (1:N).
- **User** ↔ **Wishlist** ↔ **CatalogItem**: Many-to-many.
- **LendingTransaction** ↔ **Fine**: One transaction may generate one fine (1:1).

- **User ↔ Comment**: A user posts many comments; polymorphic over archive items, research outputs, and projects; threading via self-referencing parent_id (1:N).
- **User ↔ Reaction**: A user adds reactions to items or comments; unique per (user, entity_type, entity_id, reaction_type).
- **User ↔ Announcement**: Admin creates many broadcast announcements; target_role determines recipients (1:N).
- **Event ↔ EventMaterial**: One event has many attached files/slides (1:N).
- **Event ↔ EventRSVP ↔ User**: Many-to-many with FIFO waitlist ordering.
- **User ↔ AccessRequest ↔ ArchiveItem**: A member submits access requests for Restricted-tier items; archivist reviews (1:N).

8 UI and UX Requirements

8.1 Wireframes

Below are textual wireframe descriptions for the 8 primary screens. *Convert to visual wireframes using Figma/Balsamiq for final submission.*

8.1.1 Wireframe 1: Public Archive Search & Browse

```
+-------------------------------------------------------------+
| [Logo]  Digital Knowledge Platform  [Search Bar]       [Login] |
+-------------------------------------------------------------+
| Home > Archive                                               |
+--------------+----------------------------------------------+
| FILTERS      | RESULTS (showing 1-20 of 347)                |
|              |                                               |
| Category     | +---------------------------------------------+|
| [v Select]   | | Title (EN) / Title (BN)                     ||
|              | | Author: Name | Date: 2026-01-15             ||
| Date Range   | | Tags: [Tag1] [Tag2] | Type: PDF              ||
| [From] [To]  | | Access: Public                              ||
|              | +---------------------------------------------+|
| Media Type   |                                               |
| [x] PDF      | +---------------------------------------------+|
| [x] Image    | | Another Document Title                      ||
| [ ] Audio    | | ...                                         ||
| [ ] Video    | +---------------------------------------------+|
|              |                                               |
| Language     | [< 1  2  3  4  5 ... >]                       |
| (o) All      |                                               |
| ( ) English  |                                               |
| ( ) Bangla   |                                               |
+--------------+----------------------------------------------+
```

8.1.2 Wireframe 2: Archive Item Detail & Media Viewer

```
+-------------------------------------------------------------+
```

```
| [Logo]  DKP                            [Search]        [User]  |
+-------------------------------------------------------------+
| Home > Archive > Item Title                                    |
+-------------------------------------------------------------+
| Title (EN): Introduction to Machine Learning                   |
| Title (BN): (Bangla Title)                                     |
|                                                                |
| +----------------------------+  Metadata Panel                |
| |                            |  Author: Dr. Rahman            |
| | [Embedded PDF/Media Viewer] |  Date: 2026-01-10             |
| |                            |  Category: AI/ML               |
| | Page 1 of 45  [<] [>] [+]  |  Tags: [ML] [AI]              |
| |                            |  Language: English              |
| +----------------------------+  Access: Public                |
|                                 Version: 3.0                   |
| [Download]  [Cite]  [Share]     Status: Published              |
|                                                                |
| Version History                                                |
| v3.0 - 2026-01-10 - Dr. Rahman - "Updated conclusions"        |
| v2.0 - 2025-11-05 - Dr. Rahman - "Added chapter 4"            |
| v1.0 - 2025-09-01 - Dr. Rahman - "Initial upload"             |
+-------------------------------------------------------------+
```

8.1.3 Wireframe 3: Student Project Showcase Gallery

```
+-------------------------------------------------------------+
| [Logo]  DKP                            [Search]        [Login] |
+-------------------------------------------------------------+
| Home > Student Showcase                                        |
| Filters: [Department v] [Semester v] [Technology v] [Advisor v]|
+-------------------------------------------------------------+
| +--------------+ +--------------+ +--------------+            |
| | [Thumbnail]  | | [Thumbnail]  | | [Thumbnail]  |            |
| | Project A    | | Project B    | | Project C    |            |
| | Team: 3 mem  | | Team: 4 mem  | | Team: 2 mem  |            |
| | CSE, Sp 2026 | | EEE, Fa 2025 | | CSE, Sp 2026 |            |
| | [React] [ML] | | [IoT] [C++] | | [fastapi]     |            |
| +--------------+ +--------------+ +--------------+            |
|                                                                |
| [< 1  2  3  >]                                                |
+-------------------------------------------------------------+
```

8.1.4 Wireframe 4: Library Catalog Search (Public)

```
+-------------------------------------------------------------+
| [Logo]  DKP                     [Catalog Search]        [Login] |
+-------------------------------------------------------------+
| Home > Library Catalog                                         |
+--------------+----------------------------------------------+
```

```
| FACETS       | Results for "data structures" (42 found)        |
|              |                                                 |
| Author       | +-----------------------------------------------+|
| [Search...]  | | Data Structures & Algorithms in Java         ||
| [ ] Cormen   | | Author: Goodrich | Year: 2024               ||
| [ ] Goodrich | | Category: Textbook | ISBN: 978-1-...         ||
|              | | Status: Available (3 of 5 copies)            ||
| Category     | | Shelf: CS-302-A                              ||
| [x] Textbook | +-----------------------------------------------+|
| [ ] Reference|                                                 |
|              | +-----------------------------------------------+|
| Year Range   | | Introduction to Algorithms (CLRS)            ||
| [2020][2026] | | Author: Cormen et al. | Year: 2022          ||
|              | | Status: All copies on loan                   ||
| Availability | | [Place Hold]                                 ||
| (o) All      | +-----------------------------------------------+|
| ( ) Available|                                                 |
| ( ) On Loan  | [< 1  2  3 >]                                   |
+--------------+-------------------------------------------------+
```

8.1.5 Wireframe 5: Member Dashboard

```
+-----------------------------------------------------------------+
| [Logo]  DKP                                [Bell 3] [Profile]  |
+-----------------------------------------------------------------+
| My Dashboard                                                    |
+--------------+--------------------------------------------------+
| SIDEBAR      | Welcome, Nuruzzaman!                             |
|              |                                                  |
| Active       | +- Active Borrows (2) ------------------------+ |
|   Borrows    | | Data Structures       Due: 2026-03-15       | |
| History      | |   [Renew]                                   | |
| Fines        | | Database Systems      Due: 2026-03-10 (!)   | |
| Wishlist     | |   OVERDUE - Fine: Tk 25.00                  | |
| Holds        | +---------------------------------------------+ |
| Settings     |                                                  |
|              | +- Notifications ---------------------------+ |
|              | | "Database Systems" is overdue!             | |
|              | | Your hold on "AI: Modern Approach"         | |
|              | |   is now available. Pick up by 2026-03-12  | |
|              | +---------------------------------------------+ |
|              |                                                  |
|              | +- Quick Stats ----------------------------+ |
|              | | Books Borrowed (all time): 27            | |
|              | | Active Borrows: 2                        | |
|              | | Outstanding Fines: Tk 25.00              | |
|              | | Wishlist Items: 5                         | |
|              | +-------------------------------------------+ |
```

```
+--------------+-------------------------------------------------+
```

8.1.6 Wireframe 6: Librarian Operations Dashboard

```
+-----------------------------------------------------------------+
| [Logo]  DKP -- Librarian Dashboard          [Bell] [Admin]     |
+-----------------------------------------------------------------+
| +--------+ +--------+ +--------+ +--------+ +--------+         |
| |On Loan | |Overdue | |Returns | | Holds  | | Fines  |         |
| |  142   | |   18   | |  27    | |   34   | |Tk 4500 |         |
| | today  | | items  | | today  | |pending | |pending |         |
| +--------+ +--------+ +--------+ +--------+ +--------+         |
+-----------------------------------------------------------------+
| Quick Actions: [Scan Barcode] [+ New Item] [Import]             |
+-----------------------------------------------------------------+
| Recent Transactions                                              |
| +----------+----------+------------+----------+----------+      |
| | Time     | Member   | Item       | Action   | Due Date |      |
| +----------+----------+------------+----------+----------+      |
| | 10:32 AM | ID-2045  | CLRS 4th   | Issue    | Mar 21   |      |
| | 10:28 AM | ID-1893  | OS Concepts| Return   | -        |      |
| | 10:15 AM | ID-2102  | Clean Code | Issue    | Mar 21   |      |
| +----------+----------+------------+----------+----------+      |
|                                                                  |
| [Generate Report]  [View All Overdue]  [Catalog Mgmt]           |
+-----------------------------------------------------------------+
```

8.1.7 Wireframe 7: Archivist Bulk Upload Interface

```
+-----------------------------------------------------------------+
| [Logo]  DKP -- Archive Management           [Bell] [Staff]     |
+-----------------------------------------------------------------+
| Archive > Bulk Upload                                            |
+-----------------------------------------------------------------+
| +-------------------------------------------------------------+ |
| |                                                             | |
| |   Drag & drop files here or [Browse Files]                 | |
| |   Supports: PDF, DOCX, JPEG, PNG, MP3, MP4                 | |
| |   Max 50 files, 500 MB each                                | |
| |                                                             | |
| +-------------------------------------------------------------+ |
|                                                                  |
| Selected Files (12):                                             |
| +----+--------------------+-------+----------+-----------+      |
| | OK | report_2025_q4.pdf | 12 MB | ######## | Complete  |      |
| | OK | photo_lab_01.jpeg  | 4 MB  | ######## | Complete  |      |
| | .. | thesis_draft.pdf   | 85 MB | ####.... | 52%       |      |
| | -- | interview.mp3      |120 MB | ........ | Queued    |      |
| | XX | corrupt_file.xyz   | 2 MB  |          | Invalid   |      |
```

```
| +----+--------------------+-------+----------+-----------+     |
|                                                                 |
| Default Metadata:                                               |
| Category: [v Select]  Access: [v Member]  Tags: [Add...]        |
|                                                                 |
| [Apply Defaults & Upload Remaining]  [Cancel All]               |
+-----------------------------------------------------------------+
```

8.1.8 Wireframe 8: Research Output Detail & Citation Export

```
+-----------------------------------------------------------------+
| [Logo]  DKP -- Research Repository   [Search]         [User]  |
+-----------------------------------------------------------------+
| Research > Paper Detail                                         |
+-----------------------------------------------------------------+
| Title: Deep Learning Approaches for Bangla OCR                  |
| Authors: Dr. A. Rahman, S. Hossain, M. Karim                   |
| Lab: NLP & Computer Vision Lab                                  |
| Published: 2026-01-20 | Journal: IJCV Vol. 34                 |
| DOI: DKP-2026-00042                                            |
| Keywords: [Deep Learning] [OCR] [Bangla] [NLP]                 |
|                                                                 |
| Abstract:                                                       |
| This paper presents a novel approach to Bangla OCR using        |
| transformer-based architectures...                              |
|                                                                 |
| +-----------------------------------------+                     |
| |     [Embedded PDF Viewer]                |                     |
| |     Page 1 of 12                         |                     |
| +-----------------------------------------+                     |
|                                                                 |
| [Download PDF]  [Cite v]                                        |
|                  +- BibTeX                                      |
|                  +- APA                                         |
|                  +- MLA                                         |
|                                                                 |
| Related Outputs from NLP & CV Lab:                              |
| - Bangla Handwriting Recognition (2025)                         |
| - Sentiment Analysis of Bangla Text (2025)                      |
+-----------------------------------------------------------------+
```

8.1.9 Wireframe 9: Notification Feed

```
+-----------------------------------------------------------------+
| [Logo]  DKP                          [Bell 5] [Profile]        |
+-----------------------------------------------------------------+
| Home > Notifications                 [ Mark All Read ]         |
+-----------------------------------------------------------------+
| [!] UNREAD                                                     |
```

```
| Your hold on "AI: Modern Approach" is now available.          |
| Pick up by 2026-03-12.  [Go to Catalog]            -- 2h ago  |
+-------------------------------------------------------------+
| [!] UNREAD                                                    |
| Admin Announcement: System maintenance on 2026-03-10.         |
|                                                    -- 5h ago  |
+-------------------------------------------------------------+
|  READ                                                         |
| "Database Systems" is 3 days overdue. Fine is accruing.       |
|                                                    -- 1d ago  |
+-------------------------------------------------------------+
| Notification Preferences:  [Manage Preferences]               |
+-------------------------------------------------------------+
```

8.1.10 Wireframe 10: User Profile Page

```
+-------------------------------------------------------------+
| [Logo]  DKP                    [Bell] [Profile] [Logout]      |
+-------------------------------------------------------------+
| Home > My Profile                                             |
+-------------------------------------------------------------+
| +-----------+  Nuruzzaman                                     |
| | [Avatar]  |  Department: CSE  |  Role: Student Author       |
| |  [Edit]   |  Member since: Sep 2021                         |
| +-----------+  [Edit Profile]  [Account Settings]             |
|                                                               |
| Activity Summary                                              |
| Books Borrowed: 27 | Projects Submitted: 2 | Comments: 14     |
|                                                               |
| Recent Activity                                               |
| +----------+---------------------------+--------+---------+  |
| | Date     | Activity                  | Type   | Module  |  |
| +----------+---------------------------+--------+---------+  |
| | Mar 08   | Downloaded "ML Notes"     | DL     | Archive |  |
| | Mar 07   | Borrowed "Clean Code"     | Borrow | Library |  |
| | Mar 06   | Commented on Paper XYZ    | Comment| Research|  |
| | Mar 05   | Submitted Project: DKP    | Upload | Showcase|  |
| +----------+---------------------------+--------+---------+  |
+-------------------------------------------------------------+
```

8.1.11 Wireframe 11: Analytics Dashboard (Admin/Librarian)

```
+-------------------------------------------------------------+
| [Logo]  DKP -- Analytics Dashboard         [Bell] [Admin]    |
+-------------------------------------------------------------+
| Date Range: [2026-02-01] to [2026-03-08]  [Apply]            |
+-------------------------------------------------------------+
| +----------+ +----------+ +-----------+ +----------+          |
| | Uploads  | |Downloads | | Searches  | |Active    |          |
```

```
| |  1,284   | |  8,420   | |  23,501   | |Users: 317|          |
| | +12% MoM | | +8% MoM  | | +15% MoM  | |          |          |
| +----------+ +----------+ +-----------+ +----------+          |
|                                                               |
| Daily Download Trend (bar chart)                              |
| |-----------------------------------------------------------| |
| | ##  ## ### ###  ## #### ##### ######## #### ### ####  | |
| |-----------------------------------------------------------| |
|  Feb 01              Feb 15                       Mar 08      |
|                                                                |
| Most Accessed Resources         Top Search Terms              |
| 1. Intro to ML  (428 DLs)       1. "algorithms"    (321)     |
| 2. OS Design    (312 DLs)       2. "data science"  (187)     |
| 3. DKP Showcase (289 views)     3. Zero-results: 45 queries   |
|                                                                |
| [Export PDF]  [Export CSV]                                    |
+---------------------------------------------------------------+
```

8.1.12 Wireframe 12: Resource Comments & Discussion

```
+---------------------------------------------------------------+
| [Logo]  DKP -- Archive Item              [Bell] [User]        |
+---------------------------------------------------------------+
| [Embedded Viewer / document content above]                    |
+---------------------------------------------------------------+
| [Download]  [Cite]  [Share]  [Like 34]  [Helpful 12]          |
+---------------------------------------------------------------+
| Discussion (7 comments)                    [Ask a Question]   |
|                                                                |
| +- Dr. Rahman (Author)  Mar 5, 2026 ----------------------------+
| | Thank you for the interest. Chapter 3 has the proofs.       |
| |  [Reply]  [Like 4]                                          |
| | +- Nuruzzaman  Mar 6, 2026 --------------------------------+ |
| |   | Follow-up: Does it cover LSTM variants?               | |
| |   | [Reply]  [Like 1]                                      | |
| +-----------------------------------------------------------+ |
|                                                                |
| Q&A (2 questions)                                              |
| +-----------------------------------------------------------+ |
| | Q: Is this paper peer-reviewed?  -- Sifat  Mar 4, 2026    | |
| | [v Answer] Yes, published in IJCV 2026.  -- Dr. Rahman    | |
| +-----------------------------------------------------------+ |
|                                                                |
| Add a comment: [_________________________]  [Post]            |
+---------------------------------------------------------------+
```

8.1.13 Wireframe 13: Event Registry & Calendar

```
+---------------------------------------------------------------+
```

```
| [Logo]  DKP -- Events & Seminars         [Bell] [User]        |
+-------------------------------------------------------------+
| Home > Events                       [Staff: + Create Event]    |
+------------------------------+---------------------------------+
| MARCH 2026                   | Upcoming Events                 |
| Mo Tu We Th Fr Sa Su         |                                 |
|  2  3  4  5  6  7  8        | +----------------------------+ |
|  9 10 11 12 13 14 15        | | Machine Learning Seminar    | |
| 16 17[18]19 20 21 22        | | CSE Dept  |  Mar 18, 2026   | |
| 23 24 25 26 27 28 29        | | 2:00 PM   |  Room CS-301    | |
| 30 31                        | | 12 of 40 seats remaining    | |
|                              | | [RSVP Now]   [Materials]    | |
| Selected: Mar 18             | +----------------------------+ |
|  * ML Seminar 2:00 PM        |                                 |
|  * Lab Open Day 4:00 PM      | +----------------------------+ |
|                              | | Lab Open Day                | |
|                              | | Research Lab | Mar 18       | |
|                              | | 4:00 PM | Lab Building B   | |
|                              | | Seats: Unlimited            | |
|                              | | [RSVP Now]   [Materials]    | |
|                              | +----------------------------+ |
+------------------------------+---------------------------------+
```

8.2 UX and Accessibility Baseline

Requirement	Details
Clear Navigation & Labels	Persistent top navbar with breadcrumbs on every page; sidebar navigation for dashboards; all labels in English with optional Bangla toggle.
Keyboard Accessibility	All interactive elements reachable via Tab key; focus indicators clearly visible; modal dialogs trap focus.
Readable Contrast & Typography	Minimum WCAG 2.1 AA contrast ratio (4.5:1 normal text, 3:1 large text); primary font: Inter/Noto Sans (Bangla support); base size 16px; line height 1.5.
Responsive Behavior	Mobile-first; breakpoints at 480px, 768px, 1024px; sidebar collapses to hamburger on mobile; tables become cards on small screens.
Error Handling UX	Inline form validation errors; toast notifications for success/failure; loading spinners and skeleton screens.
Bilingual Support	Language switcher (EN/BN) in navbar; all UI labels, errors, notifications in both languages.
Screen Reader Support	Semantic HTML5 elements; ARIA labels on icons; alt text on all images.

9 Non-Functional Requirements

NFR ID format: `NFR-TXX-###`

NFR ID	Category	Pri.	Requirement	Target / Metric
NFR-TXX-001	Performance	M	Page load time for search results	$\leq$ 2s for 95th percentile on 4G
NFR-TXX-002	Performance	M	File upload throughput	500 MB within 5 min on 50 Mbps
NFR-TXX-003	Performance	M	Concurrent users	500 concurrent users without degradation (p95 response time $\leq$ 2 s under catalog search load; tested with k6 or Locust; hardware baseline and ramp-up duration documented in test report)
NFR-TXX-004	Performance	M	Search response time (English)	$\leq$ 1s for 100K indexed documents
NFR-TXX-005	Scalability	S	Data growth support	1M archive items, 50K catalog records
NFR-TXX-006	Security	M	Authentication	JWT with refresh tokens; 30-min inactivity timeout
NFR-TXX-007	Security	M	Password policy	Min 8 chars, 1 upper, 1 lower, 1 digit, 1 special; bcrypt
NFR-TXX-008	Security	M	Authorization (RBAC)	RBAC enforced at API middleware and UI levels
NFR-TXX-009	Security	M	Data encryption	HTTPS TLS 1.2+; sensitive data encrypted at rest
NFR-TXX-010	Security	M	Input validation	Sanitized against XSS, SQL injection, path traversal

NFR ID	Category	Pri.	Requirement	Target / Metric
NFR-TXX-011	Security	M	File upload security	Malware scan; executables rejected; MIME + magic bytes
NFR-TXX-012	Availability	M	Uptime	99.5% (excl. scheduled maintenance)
NFR-TXX-013	Reliability	M	Backup / Disaster Recovery	Daily DB backups, 30-day retention; minimum 2-zone storage replication; automated failover; RTO $\leq$ 4 hours, RPO $\leq$ 1 hour
NFR-TXX-014	Usability	M	Learnability	Catalog search in 2 min (new user); issue/return in 1 min
NFR-TXX-015	Usability	M	Error recovery	Confirmation dialogs; soft-delete with 30-day recovery
NFR-TXX-016	Maintainability	S	Code quality	Style guide; $\geq$70% coverage; CI/CD enforces lint+test
NFR-TXX-017	Portability	M	Browser support	Chrome 90+, Firefox 88+, Edge 90+, Safari 14+
NFR-TXX-018	Compliance	M	Data privacy	Institutional policy; personal data deletable on request
NFR-TXX-019	i18n	M	Bangla language support	Full Unicode; Bangla collation in DB; Bangla metadata search (Must Have); ICU Bangla full-text analyzer (Should Have)

NFR ID	Category	Pri.	Requirement	Target / Metric
NFR-TXX-020	Monitoring	S	Observability	JSON structured logging; Prometheus metrics; email/Slack alerts
NFR-TXX-021	Performance	S	Notification delivery latency	In-app notifications within 5 s; email digests within 10 min of scheduled time
NFR-TXX-022	Performance	S	Analytics dashboard load time	Dashboard renders within 4 s with up to 1M records; export within 30 s
NFR-TXX-023	Security	S	Comment moderation	XSS-sanitized; flagged content queued within 1 s; moderator review within 24 h SLA

10 Risks and Open Questions

10.1 Risk Register

Risk ID	Description	Prob.	Impact	Mitigation
RISK-TXX-001	Bangla search tokenization may be inaccurate with default Elasticsearch analyzers; English analyzer must also be verified for academic vocabulary.	Medium	High	Configure Bangla-specific ICU analyzer and English academic analyzer; test both with a bilingual sample corpus in Sprint 1.
RISK-TXX-002	Large file uploads (500 MB) may fail due to timeouts or memory.	Medium	Medium	Implement chunked/resumable uploads (tus protocol); set server limits; retry mechanism.

Risk ID	Description	Prob.	Impact	Mitigation
RISK-TXX-003	Single point of failure: a single-server deployment risks total data loss and extended downtime if the server fails.	Medium	Critical	Minimum 2-zone storage replication; automated failover; daily DB backups with 30-day retention; RTO $\leq$ 4 hours, RPO $\leq$ 1 hour (see NFR-TXX-013).
RISK-TXX-004	Team inexperience with Elasticsearch and barcode libraries.	Medium	High	Tech spikes in Sprint 1; pair programming; learning resources.
RISK-TXX-005	Large scope (4 modules) risks incomplete delivery in 14 weeks.	High	High	Strict MoSCoW; Archive+Library core in Sprints 1–2; Research+Showcase in Sprint 3.
RISK-TXX-006	Lack of real librarian input may lead to incorrect workflow assumptions.	Medium	Medium	Schedule 2+ interviews with library staff by Week 4; validate wireframes.
RISK-TXX-007	Dirty/inconsistent data in migration spreadsheets.	Medium	Medium	Robust CSV import with per-row validation and duplicate detection.
RISK-TXX-008	Access tier bugs could expose restricted content.	Low	Critical	Access checks at API middleware + DB query levels; security test cases; peer review.
RISK-TXX-009	Transactional email delivery (digests, reminders) may be flagged as spam or delayed by institutional mail servers.	Medium	Medium	Use dedicated email service (SendGrid/SES); configure SPF, DKIM, DMARC; test with institutional domain early in Sprint 2.
RISK-TXX-010	Analytics event-log volume (clicks, searches, downloads) may grow rapidly, degrading query performance.	Medium	Medium	Partition analytics tables by month; use read replica for dashboard queries; pre-aggregate nightly summary tables.

10.2 Open Questions

#	Question	Owner	Status
1	What is the standard loan period and max borrow limit? (Proposed: 14 days, max 5)	Librarian	Open
2	What is the daily fine rate? (Proposed: Tk 5/day)	Librarian	Open
3	Should the platform support SSO (Google Workspace, LDAP)?	IT Dept.	Open
4	Are there existing catalog records in MARC or CSV to migrate?	Library Dept.	Open
5	Maximum acceptable storage budget for year 1?	Administration	Open
6	Should showcase require minimum grade for publication?	Faculty	Open
7	Preferred Bangla font or typography guideline?	Branding	Open
8	Should fines block all borrowing or only above a threshold?	Librarian	Open

11 Testability and Traceability

11.1 Requirement Traceability

Requirement ID	Story / Use Case	Planned Test ID
FR-TXX-001	US-TXX-002, UC-1	TC-TXX-001
FR-TXX-002	US-TXX-002, UC-1	TC-TXX-002
FR-TXX-003	US-TXX-002, UC-1	TC-TXX-003
FR-TXX-004	US-TXX-001, US-TXX-017, UC-5	TC-TXX-004, TC-TXX-005
FR-TXX-005	US-TXX-001, UC-5	TC-TXX-005
FR-TXX-006	US-TXX-018	TC-TXX-006
FR-TXX-007	US-TXX-012	TC-TXX-007
FR-TXX-007A	US-TXX-029	TC-TXX-029
FR-TXX-009	US-TXX-002	TC-TXX-008
FR-TXX-019	US-TXX-004, UC-3	TC-TXX-009
FR-TXX-020	US-TXX-005, UC-3	TC-TXX-010
FR-TXX-025	US-TXX-016	TC-TXX-011
FR-TXX-026	US-TXX-006, UC-4	TC-TXX-012
FR-TXX-028	US-TXX-006, UC-2	TC-TXX-013

Requirement ID	Story / Use Case	Planned Test ID
FR-TXX-029 FR-TXX-031 FR-TXX-033 FR-TXX-036 FR-TXX-038	US-TXX-008, UC-2 US-TXX-007 US-TXX-008 US-TXX-010 US-TXX-009	TC-TXX-014 TC-TXX-015 TC-TXX-016 TC-TXX-017 TC-TXX-018
FR-TXX-041	US-TXX-019	TC-TXX-019
FR-TXX-042	US-TXX-020	TC-TXX-019
FR-TXX-044	US-TXX-020	TC-TXX-020
FR-TXX-045	US-TXX-021	TC-TXX-021
FR-TXX-046	US-TXX-022	TC-TXX-022
FR-TXX-048	US-TXX-022	TC-TXX-022
FR-TXX-050	US-TXX-023	TC-TXX-023
FR-TXX-053	US-TXX-024	TC-TXX-024
FR-TXX-054	US-TXX-024	TC-TXX-024
FR-TXX-055	US-TXX-025	TC-TXX-025
FR-TXX-057	US-TXX-026	TC-TXX-026
FR-TXX-059	US-TXX-027	TC-TXX-027
FR-TXX-060	US-TXX-027	TC-TXX-027
FR-TXX-061	US-TXX-028	TC-TXX-028
FR-TXX-063	US-TXX-027, US-TXX-028	TC-TXX-027, TC-TXX-028

11.2 Initial Manual Test Plan

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-TXX-001	Upload valid PDF	Archivist logged in; PDF $\leq$500 MB	1. Go to Upload. 2. Select PDF. 3. Wait. 4. Fill metadata. 5. Save Draft.	File uploaded; metadata saved; Draft status.
TC-TXX-002	Reject unsupported file	Archivist logged in; .exe file	1. Go to Upload. 2. Select .exe. 3. Submit.	Error: “Unsupported format”; not uploaded.

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-TXX-003	Edit Bangla metadata	Draft archive item exists	1. Open item. 2. Edit title_bn. 3. Add Bangla description. 4. Save.	Bangla text saved and renders correctly.
TC-TXX-004	Bangla archive search	Published item with Bangla metadata	1. Navigate to search. 2. Enter Bangla keyword. 3. Search.	Relevant Bangla results returned.
TC-TXX-005	Multi-filter browse	20+ published items	1. Select Category=Report. 2. Select Type=PDF. 3. Select Lang=English.	Only English PDF reports shown.
TC-TXX-006	Version lifecycle	Draft archive item	1. Change to Review. 2. Change to Published.	Status updates; searchable after Published; history recorded.
TC-TXX-007	Access tier block	Restricted item; Member login	1. Search restricted item. 2. Click result. 3. Try view/download.	"Access Denied" message shown.
TC-TXX-008	Bulk upload 10 files (incl. negative cases)	Archivist logged in; 10 files (9 valid + 1 invalid)	1. Go to Bulk Upload. 2. Select 10 files (9 valid + 1 invalid format). 3. Set defaults. 4. Upload. 5. Verify 9 succeed and 1 listed with error. 6. Attempt batch of 51 files → rejected with count error.	9 files upload; invalid file listed with error; batch >50 gracefully rejected.
TC-TXX-009	Student project submit	Student Author logged in	1. Go to Submit. 2. Fill fields. 3. Upload PDF + MP4. 4. Submit.	Saved as Pending Review; advisor notified.
TC-TXX-010	Advisor approves project	Pending project; Advisor logged in	1. Open review queue. 2. Open submission. 3. Click Approve.	Status = Published; visible in gallery.

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-TXX-011	Create catalog item	Librarian logged in	1. Go to New Item. 2. Fill title, ISBN, copies=5. 3. Save.	Item created; searchable; available=5; barcode generated.
TC-TXX-012	Public catalog search	Catalog has items	1. Open catalog (not logged in). 2. Search "algorithms." 3. Click result.	Results with availability; detail shown; borrow hidden.
TC-TXX-013	Issue book to member	Available copies; eligible member	1. Scan barcode. 2. Enter member ID. 3. Confirm issue.	Transaction recorded; copies−1; dashboard updated.
TC-TXX-014	Return overdue book	Book overdue 3 days; rate Tk 5/day	1. Scan barcode. 2. Confirm return.	Return processed; copies+1; fine Tk 15 added.
TC-TXX-015	View outstanding fines	Member has overdue fine	1. Login as member. 2. Go to Fines.	Fine details shown; total balance correct.
TC-TXX-016	Barcode scan lookup	Item has barcode	1. Open scan interface. 2. Scan barcode.	Correct item identified; detail page opens within 2s.
TC-TXX-017	Place hold on unavailable book	All copies on loan	1. Search book. 2. See "All copies on loan." 3. Click Hold.	Hold recorded; position shown; notification on return.
TC-TXX-018	Generate circulation report	Transactions exist this month	1. Go to Reports. 2. Select date range. 3. Generate. 4. Export PDF.	Report shows issues/returns/overdues; PDF accurate.
TC-TXX-019	Admin broadcasts announcement	Admin logged in	1. Go to Announcements. 2. Fill title and body. 3. Set target = All. 4. Publish.	Announcement saved; all users receive in-app notification; email sent.

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-TXX-020	Member sets notification preferences	Member logged in	1. Go to Settings → Notifications. 2. Disable email digest. 3. Save.	Preferences saved; subsequent digest emails not sent; in-app still active.
TC-TXX-021	New user registers via Google OAuth	Google account available	1. Click "Sign in with Google." 2. Authorize app. 3. Complete profile.	Profile created; JWT issued; redirected to dashboard; role = Member.
TC-TXX-022	Member views unified activity history	Member has borrows, downloads, comments	1. Go to Profile → Activity. 2. Filter by type = Download.	All downloads listed chronologically; other types hidden by filter.
TC-TXX-023	Admin views usage statistics dashboard	Analytics event log seeded with known synthetic events via test script	1. Seed analytics log with synthetic events (e.g., 50 uploads, 30 downloads). 2. Navigate to Analytics. 3. Set date range to last 30 days. 4. Check charts match seeded counts.	Charts load within 4s; displayed counts exactly match seeded event totals.
TC-TXX-024	Librarian views top search terms and zero-result queries	Search logs exist	1. Go to Analytics → Search. 2. View top 20 terms. 3. Export CSV.	Zero-result list displayed; CSV exported with correct data.
TC-TXX-025	Member posts comment on archive item	Member logged in; item published	1. Open item. 2. Scroll to Discussion. 3. Type comment. 4. Post.	Comment appears threaded; author name shown; timestamp correct.
TC-TXX-026	Staff hides inappropriate comment	Comment exists; Staff logged in	1. Open item. 2. Click flag on comment. 3. Select "Hide."	Comment hidden immediately; flagged in moderation queue.

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-TXX-027	Staff creates seminar event with slides	Staff logged in	1. Go to Events → Create. 2. Fill form. 3. Attach slides PDF. 4. Publish.	Event saved; appears in calendar; slides downloadable by members.
TC-TXX-028	Member RSVPs for event	Upcoming event with capacity 40; Member logged in	1. Open event. 2. Click RSVP. 3. Confirm.	RSVP recorded; capacity counter decremented; confirmation notification sent.
TC-TXX-029	Member requests access to restricted document	Restricted item exists; Member logged in	1. Search restricted item. 2. Click "Request Access." 3. Enter reason. 4. Submit. 5. Archivist approves request.	Request recorded; archivist receives notification; after approval, member can view/download the document; denial shows rejection message.

12 Lab 2 Submission Checklist

- **Problem statement and scope are clear.** — Section 2 defines problem, target users, goals, non-goals; Section 3 defines scope boundaries.
- **Minimum 12 user stories with acceptance criteria.** — 29 user stories provided in Section 6.1 (US-TXX-001 through US-TXX-029).
- **Functional requirements table is complete.** — 64 functional requirements (FR-TXX-001 through FR-TXX-063, plus FR-TXX-007A) across all 9 modules in Section 5.
- **Data model and data dictionary are attached.** — 22 core entities in Section 7.1; comprehensive dictionary in Section 7.2; ER diagram in Section 7.3.
- **Minimum 5 wireframes or flow screens included.** — 13 detailed wireframes in Section 8.1 covering all major user flows.
- **NFR section is completed.** — 23 non-functional requirements (NFR-TXX-001 through NFR-TXX-023) in Section 9.
- **Risks and open questions are documented.** — 11 risks and 8 open questions in Section 10.
- **Initial manual test plan (15 cases) is included.** — 29 manual test cases (TC-TXX-001 through TC-TXX-029) in Section 11.2.

13 Conclusion

13.1 Summary

This Software Requirements Specification defines the complete functional, data, UI/UX, non-functional, and testability requirements for the **Digital Knowledge Platform (DKP)** — a unified, web-based academic knowledge management system developed for the University of Dhaka, Department of Computer Science and Engineering.

The platform addresses the fragmented management of institutional knowledge assets by consolidating nine modules into a single, coherent system:

1. **Digital Archive** — Centralised, versioned, bilingual document and media repository.
2. **Research Repository** — Discoverable research output management with persistent identifiers and citation export.
3. **Student Project Showcase** — Structured submission, advisor-approval workflow, and public gallery.
4. **Digital Library & Catalog** — Full lending lifecycle, fine management, and member services (manual ID-based lending; barcode/QR operations deferred to a future release).
5. **Notification & Announcement Center** — Real-time in-app feed, email digests, and admin broadcast system.
6. **User Account & Profile Management** — OAuth and email authentication, unified activity history, and self-service account controls.
7. **Analytics & Reporting Dashboard** — Platform-wide usage statistics, engagement metrics, and exportable reports.
8. **Discussion & Comment System** — Threaded comments, Q&A, reactions, and moderated community engagement.
9. **Event & Seminar Registry** — Academic event listings, material distribution, RSVP management, and calendar view.

13.2 Expected Impact

Successful delivery of DKP is expected to produce the following outcomes for the institution:

Impact Area	Expected Outcome
Knowledge Preservation	Student projects, lab experiments, and institutional documents are permanently archived and searchable, preventing knowledge loss across academic generations.
Research Visibility	Faculty and researcher outputs gain persistent identifiers and proper citation support, improving discoverability and the institution's academic standing.
Operational Efficiency	Library operations move from manual ledgers to a fully digital lending workflow, reducing processing time per transaction from minutes to seconds.

Impact Area	Expected Outcome
Student Engagement	A structured project showcase motivates students to produce higher-quality work and provides a portfolio visible to future employers and researchers.
Community & Collaboration	Threaded discussions and Q&A on academic resources foster peer learning and direct author–reader interaction.
Data-Driven Management	Analytics dashboards give administrators and librarians evidence-based insight to improve collection growth, resource allocation, and event planning.
Inclusivity	Bilingual (Bangla and English) metadata and UI are designed from the outset; English search is delivered in v1 (**Must Have**); Bangla metadata search (title, author, tags) is **Must Have**; Bangla full-text ICU tokenization is **Should Have**.

13.3 Team Reflection

The Semicolon Squad undertook this SRS as the first formal deliverable of CSE 4113: Internet Programming Lab. This process reinforced several key software engineering lessons:

- Writing precise, testable requirements is significantly harder than writing working code; ambiguity in requirements directly translates to bugs in implementation.
- Stakeholder analysis and the MoSCoW prioritisation method are essential for managing feature scope within a fixed timeline.
- Designing a bilingual system from the outset (rather than retrofitting Bangla support later) requires deliberate decisions at every layer — database collation, search tokenisation, and UI layout.
- A well-structured data model prevents the most common class of integration defects and should be agreed upon before any coding begins.

The team is committed to implementing DKP faithfully to this specification over the remaining weeks of the semester, beginning with the Digital Archive and Digital Library core modules, and iterating through the remaining seven modules in subsequent sprints.

A Glossary of Terms

Term	Definition
Priority notation	**M** = Must Have (core MVP) **S** = Should Have **C** = Could Have (stretch goal)

Term	Definition
Access Tier	One of four permission levels applied to each resource item: Public (anyone), Member (registered and authenticated users), Staff (archivists and librarians), or Restricted (explicit grant by admin).
Archivist	A staff user role with permission to upload, edit, delete, and bulk-manage archive items and their metadata.
Audit Log	An append-only, immutable record of every create, read, update, and delete action performed on the platform, used for accountability and forensic review.
Bangla Tokenisation	The process of splitting Bangla-script text into searchable tokens using a Unicode-aware linguistic analyser (e.g., Elasticsearch ICU plug-in). Bangla metadata search (title, author, tags) is **Must Have**; full-text ICU tokenization is **Should Have**.
BibTeX	A reference management file format used by LaTeX for bibliography entries; also supported as a citation export format (alongside APA and MLA).
DOI	Digital Object Identifier — a persistent, unique handle assigned to a published research output, used as a stable citation reference.
DKP	Digital Knowledge Platform — the name of the system described by this SRS.
FIFO	First-In, First-Out — the ordering policy applied to library hold queues.
Manual Lending Entry	The book-issue and book-return method used in v1 of DKP: a librarian manually types the catalog item ID and the member ID into the lending form. This replaces barcode/QR scanning, which is excluded from v1 scope.
NAS	Network-Attached Storage — a file-level storage server connected to a local network; used in DKP as the primary file storage layer for uploaded documents, images, audio, and video.
OAuth 2.0	An open authorisation standard that allows users to authenticate via a third-party identity provider (e.g., Google) without sharing their password with DKP.
RBAC	Role-Based Access Control — a security model where permissions are assigned to roles (not individual users), and users are granted one or more roles.

Term	Definition
Soft Delete	A deletion strategy that marks a record as inactive (deleted flag = true) without physically removing it from the database, allowing recovery within a retention window.
SSO	Single Sign-On — an authentication scheme that allows a user to log in once (e.g., via the university's identity provider) and gain access to multiple systems.
TUS Protocol	An open protocol for resumable file uploads over HTTP, used to handle large file uploads (up to 500 MB) reliably without starting over after a network interruption.
Version State	The lifecycle status of an archive item or research output: Draft → Review → Published → Archived.
Waitlist	A secondary queue for RSVP registrations that exceed an event's capacity; members are promoted from waitlist to confirmed when cancellations occur.
WCAG 2.1 AA	Web Content Accessibility Guidelines version 2.1, Level AA — the accessibility standard that DKP targets for contrast ratios, keyboard navigation, and screen-reader support.
XSS	Cross-Site Scripting — a web security vulnerability where an attacker injects malicious scripts into content viewed by other users; mitigated by server-side input sanitisation.

B Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
APA	American Psychological Association (citation format)
CI/CD	Continuous Integration / Continuous Deployment
CRUD	Create, Read, Update, Delete
CSE	Computer Science and Engineering
CSV	Comma-Separated Values
DB	Database
DKP	Digital Knowledge Platform
DOI	Digital Object Identifier

Abbreviation	Full Form
ER	Entity-Relationship
FAQ	Frequently Asked Questions
FK	Foreign Key
FR	Functional Requirement
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
ICU	International Components for Unicode
ISBN	International Standard Book Number
JSON	JavaScript Object Notation
JSONB	JSON Binary (PostgreSQL native binary JSON storage)
JWT	JSON Web Token
LDAP	Lightweight Directory Access Protocol
LMS	Learning Management System
MIME	Multipurpose Internet Mail Extensions
MLA	Modern Language Association (citation format)
MoSCoW	Must / Should / Could / Won't prioritisation
MVP	Minimum Viable Product
NAS	Network-Attached Storage
NFR	Non-Functional Requirement
OGG	Ogg multimedia container format
PK	Primary Key
PPTX	PowerPoint Open XML format
RBAC	Role-Based Access Control
REST	Representational State Transfer
SRS	Software Requirements Specification
SSO	Single Sign-On
TLS	Transport Layer Security
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
US	User Story

Abbreviation	Full Form
UUID	Universally Unique Identifier
UX	User Experience
WCAG	Web Content Accessibility Guidelines
XSS	Cross-Site Scripting

C Technology Stack Summary

Layer	Technology	Purpose
Frontend	React.js / Next.js	Component-based SPA / SSR web application; responsive design
Backend API	Node.js (Express) or fastapi	RESTful API server; business logic; authentication middleware
Primary Database	PostgreSQL	Relational data: users, transactions, entities, audit logs
Search Engine	Elasticsearch + ICU	English full-text search (**Must Have**); Bangla metadata search (**Must Have**); ICU Bangla analyser for Bangla full-text tokenisation (**Should Have**)
File Storage	Server Filesystem / NAS	Local or network-attached storage for documents, images, audio, video files
Authentication	JWT + OAuth 2.0	Stateless session tokens; Google and institutional SSO login
Email Service	SendGrid / SMTP	Transactional email for digests, due-date reminders, announcements
Caching	Redis	Session data, notification counters, search result caching
Monitoring	Prometheus + Grafana	Metrics collection, dashboards, threshold-based alerting
Logging	Winston / structlog	JSON structured application logs; centralised log aggregation
CI/CD	GitHub Actions	Automated lint, test, build, and deploy pipelines
Containerisation	Docker + Docker Compose	Reproducible development and production environment packaging
File Upload	tus-js-client	Resumable chunked file upload protocol for large files

D Sprint Roadmap (Indicative)

Sprint	Weeks	Deliverables
Sprint 0	1–2	SRS finalisation; environment setup; database schema design; tech spikes (Elasticsearch Bangla analyser, file upload pipeline, OAuth flow)
Sprint 1	3–5	User Account & Profile Management (FR-TXX-045 to FR-TXX-049) + Digital Archive core (FR-TXX-001 to FR-TXX-008); basic search
Sprint 2	6–8	Digital Archive advanced: versioning lifecycle (**S**), audit trail (**S**); Digital Library & Catalog core: CRUD, search, lending/return, due-date tracking (**M**); overdue fines, dashboard, reports (**S**); bulk upload, CSV import, hold requests, QR scanning deferred as Could Have / Won't Have
Sprint 3	9–10	Research Repository (FR-TXX-013 to FR-TXX-018); Student Project Showcase (FR-TXX-019 to FR-TXX-024); advisor approval workflow
Sprint 4	11–12	Notification & Announcement Center (FR-TXX-041 to FR-TXX-044); Discussion & Comment System (FR-TXX-055 to FR-TXX-058); Analytics Dashboard (FR-TXX-050 to FR-TXX-054)
Sprint 5	13–14	Event & Seminar Registry (FR-TXX-059 to FR-TXX-063); end-to-end testing (TC-TXX-001 to TC-TXX-029); bug fixing; documentation; final submission

— *End of Document* —